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Smart Thermal System for Patient Safety Monitoring

Contact (Touch) Detection — Algorithm Evaluation

Waveshare 26984 Thermal Sensor 80×62 px 3 Synchronised Views Building on the
Human-Detection Stage

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1 Overview

This report evaluates the **contact-detection** (“touch”) stage of the Smart Thermal System — the module responsible for the *physical contact between patients and violent/sexual behaviour* scenarios — on the Waveshare 26984 dataset (three synchronised 80×62 px sensors at 8 Hz). It is a direct continuation of the *Human & Fire Detection* evaluation: two of the three contact detectors are built **on top of** the human detector characterised there, so we first recap that result (Part I) and then show how contact detection builds on it (Part II).

Three contact detectors are compared, spanning the full design spectrum of §4.4.3:

Geometric multi-view (§4.4.3.1):

person boxes from the human detector are projected onto a common floor plane through a homography and flagged as contact when two people come within a distance threshold. Deterministic, explainable, no training.

MV-STGCN (§4.4.3.2):

the same detection+projection front-end feeds a Multi-View Spatiotemporal Graph CNN that learns contact over a 16-frame window of actor graphs.

Thermo-X3D (§4.4.3.3):

a pixel-based (2+1)D convolutional network over the raw thermal video volume. It uses *no* bounding boxes and *no* homography — the designed failsafe for when close contact merges two bodies into a single thermal blob.

Headline result. The pixel-based Thermo-X3D is the clear winner — best F1 *and* the lowest false-alarm rate by a wide margin — confirming the §4.4.3.3 design hypothesis that a method independent of the detector and the homography is the robust one. The two front-end-dependent methods inherit the limitations of their inputs: a self-calibrated homography that is accurate enough to localise but too noisy to fuse cleanly, and the merging of two bodies into one detection at the moment of contact.

1.1 Common methodology

As in the human/fire evaluation, every frame is reduced to a binary verdict (contact present / absent) and scored with a confusion matrix and four derived metrics:

$$\begin{aligned} \text{Accuracy} &= \frac{TP+TN}{TP+TN+FP+FN}, & \text{Precision} &= \frac{TP}{TP+FP}, \\ \text{Recall} &= \frac{TP}{TP+FN}, & \text{F1} &= \frac{2P \cdot R}{P+R}. \end{aligned}$$

Recall is the safety metric (a missed assault is a direct failure); **precision** and the false-alarm rate guard against alert fatigue, which in a psychiatric ward is itself a safety risk.

Part I — The Human-Detection Foundation

2 Recap: which human detector feeds contact

The geometric and MV-STGCN contact detectors begin with per-camera person detection. From the human-detection evaluation, **MobileNet-SSD** is the production choice on the Waveshare sensor; Table 1 reproduces its test-split result alongside the two classical baselines.

Table 1: Human detection — aggregate test-split metrics under the strict localisation criterion ($\text{IoU} \geq 0.5$), from the Human & Fire evaluation. MobileNet-SSD is used as the person detector for the contact front-ends.

Detector	Acc	Prec	Rec	F1	Mean IoU	False-alarm
AdaptiveThreshold	76.6%	93.4%	80.7%	86.6%	56.5%	83.0%
HOG+SVM	79.8%	92.7%	85.1%	88.7%	58.5%	98.1%
MobileNet-SSD	95.8%	99.1%	96.4%	97.7%	67.3%	13.2%

Two properties of MobileNet-SSD matter for contact detection:

- **It rarely misses people.** 96.4 % recall, rising to a perfect 100 % F1 on the crowded scenes (3ppl_dance, 3pp_surprise, 3pplhedroncolider, 2ppl_fight). On the contact-relevant frames specifically, it finds at least two separate people in **96 % of true-contact frames** — so a failure to flag contact is almost never a failure to *see* the people.
- **Its boxes are well-localised.** 67.3 % mean IoU at 80×62 yields a foot-point stable enough to project — the prerequisite for the geometric and MV-STGCN front-ends, which the 32×24 MLX90640 could not provide.

The central finding of Part II is therefore that **good human detection is necessary but not sufficient for contact detection**: the residual difficulty lies downstream, in the multi-view geometry and in the physics of two bodies touching.

Part II — Contact Detection

3 Setup (Contact Detection)

3.1 Labels and the data-scarcity reality

Contact is annotated as a per-frame binary label in `contact_labels.csv` (`session`, `frame_idx`, `contact`). Across the dataset only **188 frames are labelled contact-positive**, concentrated in four scenes: 3pp_surprise (66), 3pplhedroncolider (53), 2ppl_fight (50) and 2ppl_hug (19). Every other scene — single-person, walking, cigarette, heater, empty — supplies **contact-negative** frames, including the crucial *hard negatives* 2ppl_walk and the_more_the_merrier, in which several people are present and near one another but not in contact.

Table 2: Contact label balance. Positives are rare ($\approx 7\%$ of labelled frames) and confined to four scenes — a severe constraint on the learned detectors, addressed below with class-weighted training.

	Frames	Share
Contact-positive	188	6.8 %
Contact-negative	2,565	93.2 %
Total labelled	2,753	100 %

3.2 Self-calibrated homography (no floor coordinates)

The geometric and MV-STGCN detectors need a homography H_k mapping each camera’s image plane to a common floor plane. Classical hot-point calibration requires heated targets at *known* floor coordinates, which were not recorded for this dataset. We instead **self-calibrate from the calibration walk**: in `calibrate_room` a single person is visible to all three cameras simultaneously, so that person’s foot-point in each view is an image of *one common floor point*. Across the 146-frame track this yields, for each camera pair, a set of image-to-image correspondences induced purely by the floor plane. Taking camera 0 as the reference plane ($H_0 = I$), we solve $H_{1 \rightarrow 0}$ and $H_{2 \rightarrow 0}$ by DLT inside a RANSAC loop.

Table 3: Self-calibration quality. RANSAC consensus is modest — only $\approx 25\text{--}30\%$ of frames are inliers — because a standing person’s bounding-box foot-point is not a perfectly stable floor point across viewing angles. The fitted inliers are tight ($\leq 1.7\text{ px}$), but the low inlier ratio foreshadows the fusion difficulty in §5.

Camera	Role	Corr. frames	RANSAC inliers	Median residual
0	reference ($H_0 = I$)	—	—	—
1	→ cam-0 plane	123	36	1.38 px
2	→ cam-0 plane	146	30	1.66 px

Consequence (important). Because the reference plane is camera 0’s *image*, distances are measured in camera-0 floor-pixels, not metres, and are perspective-distorted. The clustering radius ε and contact threshold δ are therefore tuned empirically rather than set to physical values; the operating point used below is $\varepsilon = 15\text{ px}$, $\delta = 18\text{ px}$.

3.3 Detectors and decision rule

1. **Geometric** — foot-point projection → cross-camera clustering/validation (single-source clusters discarded) → contact if any actor pair is within δ . No training.
2. **MV-STGCN** — MobileNet-SSD detections → Kalman tracking (I²C phase-shift correction) → homographic fusion to actor nodes → adaptive-adjacency ST-GCN over a $T = 16$ window → 2-class softmax. Trained (class-weighted cross-entropy).
3. **Thermo-X3D** — per-camera (2+1)D X3D encoder over the raw 16-frame thermal volume with CBAM thermal attention, late-fused across the three views → 2-class softmax. No detections, no homography. Trained (class-weighted cross-entropy).

For the learned detectors the raw per-frame contact probability is captured and the decision threshold is swept on a validation split (maximising F1) before being applied to the test split.

3.4 Evaluation protocol

The temporal models (MV-STGCN, Thermo-X3D) carry a 16-frame rolling buffer and therefore require *contiguous* frames. Each session’s timeline is split without shuffling: first 60 % → train, next 15 % → validation, final 25 % → test; at evaluation the detector is reset per session and the whole session is replayed in order so the buffer is warm before the scored frames. **All three detectors are scored on this single common test set of 695 frames (41 contact-positive, 654 negative)** — the rule-based Geometric detector, which needs no training, is evaluated on the same test segments (its train portion is simply ignored), so the comparison below is frame-for-frame identical across the three methods.

4 Results (Contact Detection)

4.1 Aggregate

Table 4: Contact detection — aggregate metrics on the common 695-frame test set (41 contact-positive, 654 negative); all three detectors scored frame-for-frame under the identical timeline protocol (§3.4). **Thermo-X3D leads on F1 and on false-alarm rate simultaneously.**

Detector	Front-end	Acc	Prec	Rec	F1	False-alarm
Geometric	SSD + homography	91.4%	26.8%	26.8%	26.8%	4.6%
MV-STGCN	SSD + homography + GCN	70.9%	14.8%	82.9%	25.2%	29.8%
Thermo-X3D	raw pixels	94.0%	48.4%	36.6%	41.7%	2.4%

Geometric confusion (test, 41 contact / 654 no-contact).

	Pred: Contact	Pred: None
Truth: Contact (41 fr.)	TP = 11	FN = 30
Truth: None (654 fr.)	FP = 30	TN = 624

MV-STGCN confusion (test, 41 contact / 654 no-contact).

	Pred: Contact	Pred: None
Truth: Contact (41 fr.)	TP = 34	FN = 7
Truth: None (654 fr.)	FP = 195	TN = 459

Thermo-X3D confusion (test, 41 contact / 654 no-contact).

	Pred: Contact	Pred: None
Truth: Contact (41 fr.)	TP = 15	FN = 26
Truth: None (654 fr.)	FP = 16	TN = 638

4.2 Per-scene breakdown

The three detectors fail in characteristically different ways, which the per-scene view makes explicit (Table 5). The first three rows are the contact-bearing test scenes (recall is what matters); the remainder are pure hard negatives where every prediction is a false alarm. Reading down the false-alarm columns is the clearest summary: Thermo-X3D raises *zero* false alarms on every single-person or walking scene, the Geometric detector scatters a few, and MV-STGCN fires on essentially all of them.

Table 5: Per-scene test results for all three detectors (common 695-frame test set). “Pos” is the number of contact-positive frames in that scene’s test segment; “R” is recall on those, “FP” the false-alarm count. Totals: Geometric 30 FP, Thermo-X3D 16 FP, MV-STGCN 195 FP.

Scene	Pos	Geometric		Thermo-X3D		MV-STGCN	
		R	FP	R	FP	R	FP
2ppl_fight	19	0%	1	0%	0	100%	11
3pp_surprise	15	67%	2	100%	3	100%	3
3pplhedroncolider	7	14%	3	0%	13	0%	0
3ppl_dance	0	—	11	—	0	—	34
the_more_the_merrier	0	—	4	—	0	—	24
2ppl_walk	0	—	1	—	0	—	30
3pp_standing_still	0	—	6	—	0	—	0
2ppl_hug	0	—	1	—	0	—	14
1person cig	0	—	0	—	0	—	64
1ppl_single_forever	0	—	0	—	0	—	10
1person	0	—	0	—	0	—	5

5 Analysis

5.1 Two failure modes, and why Thermo-X3D escapes both

The contact problem has two distinct failure modes; the front-end-dependent detectors hit one or both, while the pixel-based detector is structurally immune to each.

Failure mode #1 — homography noise.

The self-calibrated homography (§3.2) localises well but fuses poorly. A diagnostic on the true-contact frames is decisive: with the strict cross-camera rule (an actor must be confirmed by ≥ 2 cameras agreeing within ε), the mean number of surviving actors per contact frame is **0.16** — the cameras almost never agree, so the geometric detector has nobody to measure. Relaxing the rule recovers actors but the *same* person now projects to several different places and registers as several phantom actors. This is exactly what makes MV-STGCN fire on non-contact crowds: in **1person cig** (one person, 64 false alarms), **2ppl_walk** (30) and **3ppl_dance** (34) the mis-projected duplicates fall within δ of one another and look like contact. Hence its 82.9% recall is bought at a 29.8% false-alarm rate. The Geometric detector suffers the same effect but more mildly: its strict two-camera validation and conservative δ leave only scattered false alarms (11 on **3ppl_dance**, 6 on **3pp_standing_still**) for a 4.6% rate — but at the price of very low recall (26.8%), since the same conservatism that suppresses phantom pairs also suppresses real ones.

Failure mode #2 — blob merge at contact.

When two people genuinely touch (a hug, a grapple) their warm bodies merge into a *single* thermal blob, so the detector returns one box and there is no pair to measure. This is intrinsic to any box-based method and is the explicit motivation for Thermo-X3D in §4.4.3.3.

Thermo-X3D uses neither boxes nor homography, so neither failure mode applies. The payoff is visible in Table 5: its false alarms are near zero on every negative scene (FAR 2.4%),

versus 4.6 % Geometric and 29.8 % MV-STGCN), and it still recovers the contact it was given enough data to learn (`3pp_surprise`, 100 % recall). Moreover its residual false alarms are confined to genuine multi-person interaction scenes (`3pp_surprise`, `3pplhedroncolider`) and never fire on a lone individual or a walking pair — a qualitatively far safer error profile than MV-STGCN flagging a single smoker (`1person cig`, 64 false alarms) as an assault. It is the only detector that is both sensitive *and* quiet.

5.2 Recall is data-limited, not method-limited

The honest qualifier mirrors the recall ceiling found for fire detection, where recall was bounded by physics rather than the algorithm. Here it is bounded by **data**: the entire dataset contains only 188 positive frames, of which ≈ 90 windows reach training. Thermo-X3D learned `3pp_surprise` perfectly but missed `2ppl_fight` in test — not because the model cannot represent a fight, but because too few fight frames were available to fit it. The per-session timeline split also lets train and test share scenes, so the 48.4 % precision, while the best of the three, should be read as a feasibility signal rather than a deployment number.

This makes the single highest-leverage next step clear: **more annotated contact recordings**, ideally with entire scenes held out of training, would directly lift Thermo-X3D’s recall without any change to the method.

6 Conclusions

- **Thermo-X3D is the contact detector to deploy.** It achieves the best F1 (41.7 %) and by far the lowest false-alarm rate (2.4 %), because being pixel-based it sidesteps both the homography-noise and blob-merge failure modes that limit the geometric and graph-based detectors.
- **Human detection is necessary but not sufficient.** MobileNet-SSD finds the people (it sees ≥ 2 separate subjects in 96 % of contact frames); the contact difficulty is entirely downstream, in multi-view geometry and in the physics of touching bodies. This is why the detector-independent method wins.
- **MV-STGCN is recall-strong but precision-poor** on this data: it recovers 82.9 % of contacts but raises false alarms on essentially every multi-person non-contact scene, a direct consequence of the self-calibrated homography. A metric homography from proper on-site hot-point calibration is the prerequisite for making the geometric and graph-based detectors viable.
- **The limiting resource is labelled contact data.** With only 188 positive frames the learned models are data-starved; expanding the contact corpus is expected to raise Thermo-X3D recall directly, and is a higher priority than further algorithmic tuning.